

Week 1 — Science: Biology

Topics: Living Things & Habitats · Plants · Animals & Humans · Life Cycles · Food Chains · Classification · Working Scientifically

KS2 Curriculum Years 3–5 · Aligned to FSCE Entrance Test format · End of session: 10 practice questions



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HOW TO USE THIS LESSON

Parent's Guide

This lesson has four parts:

- 1 Lesson Notes** — read these together or let your child read them. Key terms are highlighted.
- 2 Quick-Check Questions** — short questions after each topic. Check answers at the end of the document.
- 3 10 FSCE-Style Practice Questions** — mix of multiple choice and short written answer, just like the real test.
- 4 Answer Key** — all answers with explanations, for parent use.

Suggested timing: Read lesson notes (20 min) · Quick-check questions (10 min) · Practice questions (20 min) · Review answers together (10 min)

TOPIC 1

Living Things & Habitats

What is a living thing?

Scientists use seven life processes to check whether something is alive. You can remember them with the acronym MRS GREN:

Letter stands for...	What it means
M — Movement	Living things move (animals move around; plants move towards light)
R — Reproduction	Living things make offspring (babies, seeds, eggs)
S — Sensitivity	Living things respond to their environment (light, heat, touch)
G — Growth	Living things get bigger over their lifetime
R — Respiration	Living things release energy from food (this happens inside cells — it is NOT just breathing)
E — Excretion	Living things remove waste products from their body
N — Nutrition	Living things take in food or make their own food (plants do this through photosynthesis)

Habitats

A habitat is the natural home of a plant or animal — the place where it finds food, shelter and mates. Organisms are adapted (specially suited) to their habitat.

Key Term	Definition
Habitat	The natural environment where a plant or animal lives (e.g. woodland, pond, desert)
Microhabitat	A very small habitat within a larger one (e.g. under a log, beneath a stone)
Adaptation	A feature that makes a living thing well suited to its environment
Population	All the individuals of one species living in one area
Community	All the different populations of living things sharing one habitat
Ecosystem	A community of living things and the non-living environment they interact with

FSCE EXAM FOCUS: Applying habitat knowledge

The test won't simply ask 'what is a habitat'. It will show you a scenario and ask you to apply your knowledge. Example:

"Many trees are cut down in a rainforest where lots of animals live. What are the animals likely to need to do?"

Answer: LEAVE — the animals lose their food source and shelter, so they must leave to find a new habitat.

Food Chains & Food Webs

A food chain shows the flow of energy from one organism to the next through feeding. Every food chain starts with a producer.

Key Term	Definition
Producer	An organism that makes its own food using sunlight (always a plant or alga). Always the START of a food chain.
Consumer	An organism that eats other organisms to get energy
Primary consumer	The animal that eats the producer (usually a herbivore)
Secondary consumer	The animal that eats the primary consumer
Predator	An animal that hunts and eats other animals
Prey	An animal that is hunted and eaten by a predator
Herbivore	An animal that only eats plants
Carnivore	An animal that only eats other animals
Omnivore	An animal that eats both plants and animals
Decomposer	An organism (e.g. fungi, bacteria) that breaks down dead material and recycles nutrients

Example food chain:

Oak Tree → Caterpillar → Blue Tit → Sparrowhawk

Oak Tree = Producer **Caterpillar** = Primary consumer (herbivore) **Blue Tit** = Secondary consumer (carnivore) **Sparrowhawk** = Tertiary consumer (carnivore, top predator)

The arrows show the direction of energy transfer — energy flows FROM prey TO predator. The arrow means 'is eaten by' or 'energy flows to'.

Human Impact on Environments

Humans change environments — sometimes helping, sometimes harming living things.

Negative human impact

Positive human impact

Deforestation — cutting down forests for farming or building	Nature reserves — protected areas where wildlife can thrive
Pollution — chemicals, plastic and waste harming habitats	Garden ponds — providing microhabitats for frogs, newts, insects
Overfishing — removing too many fish from oceans	Rewilding — reintroducing species that have disappeared
Climate change — altering habitats through temperature rise	Recycling — reducing waste that harms ecosystems

TOPIC 2

Plants — Structure, Growth & Life Cycle

Parts of a flowering plant and their functions

Plant part	Function (job)
Roots	Absorb water and minerals from the soil; anchor the plant in the ground
Stem / Trunk	Transport water and nutrients between roots and leaves; support the plant
Leaves	Make food for the plant through photosynthesis (using sunlight, water and carbon dioxide)
Flowers	Reproduction — attract pollinators and produce seeds
Seeds	Contain the embryo plant; disperse to grow into new plants
Petals	Often brightly coloured to attract insects for pollination
Stamen	Male part of the flower — produces pollen
Stigma / Pistil	Female part of the flower — receives pollen during pollination

What plants need to grow

A healthy plant needs ALL of the following:

- Sunlight — for photosynthesis (making food)
- Water — absorbed through roots; used in photosynthesis
- Carbon dioxide (CO₂) — absorbed through tiny holes in leaves called stomata; used in photosynthesis
- Nutrients / minerals from soil — for healthy growth (e.g. nitrates for making proteins)
- Suitable temperature — too hot or too cold and the plant cannot function
- Space to grow — roots and leaves need room to expand

Photosynthesis word equation:

Carbon dioxide + Water → Glucose + Oxygen (requires: sunlight + chlorophyll)

Life Cycle of a Flowering Plant

The life cycle of a flowering plant follows this sequence:

**Seed → Germination → Seedling → Adult Plant → Flowering → Pollination
→ Seed Formation → Seed Dispersal → Seed**

And the cycle begins again!

Key Term	Definition
Germination	When a seed starts to grow — it needs water, warmth and oxygen (NOT light at first)
Pollination	Transfer of pollen from the stamen (male) to the stigma (female) — done by insects or wind
Fertilisation	When the pollen cell joins with the egg cell inside the flower to form a seed
Seed dispersal	How seeds travel away from the parent plant — by wind, water, animals or explosion
Asexual reproduction	Making new plants from one parent only — e.g. runners (strawberries), bulbs, tubers
Deciduous	Trees that lose their leaves in autumn (e.g. oak, sycamore, horse chestnut)
Evergreen	Trees that keep their leaves all year round (e.g. pine, holly, ivy)

How seeds are dispersed

Method of dispersal	Examples and features
Wind	Seeds are lightweight and may have wings or parachutes — e.g. dandelion (parachute), sycamore (helicopter wing), ash
Animals (external)	Seeds have hooks or sticky coatings that cling to fur or clothing — e.g. burdock, goosegrass
Animals (eaten)	Seeds inside tasty fruits — animals eat the fruit and pass seeds out in their droppings far away — e.g. berries, cherries
Water	Seeds float and are carried by rivers or the sea — e.g. coconuts, water lily
Explosion / self	Seed pods dry out and burst, firing seeds — e.g. peas, sweet peas, squirting cucumber

Animals Including Humans

Vertebrates and Invertebrates

The animal kingdom is first divided into vertebrates (animals with a backbone) and invertebrates (animals without a backbone).

Vertebrate group	Key features + examples
Fish	Breathe through gills; live in water; cold-blooded; scaly skin; lay eggs — e.g. salmon, shark, goldfish
Amphibians	Live on land AND in water; moist skin; cold-blooded; lay eggs in water — e.g. frog, toad, newt
Reptiles	Scaly skin; cold-blooded; most lay eggs on land — e.g. lizard, snake, crocodile, tortoise
Birds	Feathers; warm-blooded; two wings; lay hard-shelled eggs; beak/bill — e.g. eagle, penguin, robin
Mammals	Fur/hair; warm-blooded; give birth to live young (mostly); produce milk; breathe air — e.g. dog, whale, bat, human

Cold-blooded vs. warm-blooded — what does it mean?

Cold-blooded: body temperature changes with the environment. These animals need the sun to warm up (e.g. lizards basking on rocks).

Warm-blooded: body temperature stays roughly the same regardless of the environment. These animals generate their own heat internally (birds and mammals).

The Human Digestive System

Digestion breaks down food into small molecules that can be absorbed into the blood and used by the body.

Key Term	Definition
Mouth	Food is chewed and mixed with saliva; teeth break food into smaller pieces; saliva begins breaking down starch
Oesophagus	The tube that carries food from mouth to stomach (food travels by muscular contractions — peristalsis)
Stomach	Churns food and adds acid and enzymes to break down proteins
Small intestine	Most digestion happens here; nutrients are absorbed into the bloodstream through the walls
Large intestine	Water is absorbed back into the body; undigested food forms faeces

Rectum / Anus	Faeces stored and then removed from the body
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Types of Human Teeth

Tooth type	Function and position
Incisors (8)	Sharp, flat teeth at the front — cut and bite food
Canines (4)	Pointed teeth next to incisors — grip and tear food
Pre-molars (8)	Flat, ridged teeth — crush and grind food
Molars (12)	Large flat teeth at the back — crush and grind tough food

The Human Skeleton and Muscles

The skeleton has four main functions:

- Support — holds the body upright
- Protection — protects vital organs (skull protects the brain; ribcage protects heart and lungs)
- Movement — bones act as levers; muscles attached to bones pull them to create movement
- Blood cell production — red and white blood cells are made in bone marrow

*Muscles work in pairs (antagonistic pairs): when one contracts, the other relaxes.
Example: bicep contracts → arm bends (flexes); tricep contracts → arm straightens (extends).*

Human Development and Life Stages

Humans go through these main life stages:

Life stage	Key features
Baby / Infant	Born helpless; grows rapidly; learns to walk and talk; relies completely on adults
Child	Grows steadily; develops physical and mental skills; attends school
Adolescent (puberty)	Body changes: growth spurt, changes in body shape and reproductive development
Adult	Full physical development; can reproduce; responsible for own life
Old age	Slower movement; muscles weaken; immune system less effective; wisdom increases

TOPIC 4

Life Cycles — Comparing Mammals, Birds, Amphibians & Insects

The FSCE test may ask you to compare life cycles across different animal groups. The key differences are in how they reproduce and how their young develop.

Animal group	Life cycle summary
Mammal (e.g. human, dog, whale)	Mate → internal fertilisation → gestation (pregnancy) inside mother → live birth → parental care → juvenile → adult → mate again. Young are fed on mother's milk. Long parental care period.
Bird (e.g. robin, penguin, eagle)	Mate → internal fertilisation → lay hard-shelled eggs → incubation (sitting on eggs to keep warm) → hatching → parents feed chicks → juvenile → adult.
Amphibian (e.g. frog, toad, newt)	Mate in water → external fertilisation (eggs fertilised outside body) → eggs (spawn) → tadpole (lives in water, breathes through gills) → metamorphosis → froglet → adult frog (lives on land and water).
Insect (complete metamorphosis) (e.g. butterfly, beetle, bee)	Egg → larva (e.g. caterpillar) → pupa (chrysalis) — completely changes inside → adult (imago). This is called complete metamorphosis (4 stages).
Insect (incomplete metamorphosis) (e.g. grasshopper, dragonfly)	Egg → nymph (looks like a small adult, no wings) → adult. This is called incomplete metamorphosis (3 stages — NO pupa stage).

KEY COMPARISON — what to remember:

- Frogs: external fertilisation in water; tadpoles breathe with gills
- Butterflies: FOUR stages (egg → caterpillar → pupa → adult) — this is COMPLETE metamorphosis
- Grasshoppers: THREE stages (egg → nymph → adult) — INCOMPLETE metamorphosis (no pupa)
- Mammals: only group to feed young on milk; usually internal gestation
- Birds: only group to lay hard-shelled eggs AND have feathers

TOPIC 5

Classification of Living Things

Classification means sorting living things into groups based on shared characteristics. Scientists use a system developed by Carl Linnaeus.

The five main kingdoms

Kingdom	Examples and key features
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Animals	Cannot make own food; consume other organisms; most can move; includes vertebrates and invertebrates
Plants	Make own food through photosynthesis; cannot move; have cell walls made of cellulose
Fungi	Cannot make own food; feed on dead/decaying matter; includes mushrooms, moulds, yeast
Bacteria	Single-celled; no nucleus; very small; can be helpful (gut bacteria) or harmful (some cause disease)
Protists	Mostly single-celled; some make own food; some are animal-like — e.g. amoeba, algae

Classification keys

A classification key (also called a branching key or dichotomous key) is a tool used to identify organisms. At each step you answer YES or NO to a question about a feature, and the answers lead you to the correct name.

Example: Is it a vertebrate? YES → Does it have feathers? YES → It is a BIRD.

When using or making a key:

- Each question must have exactly two possible answers (yes/no or present/absent)
- Questions should be based on observable features, not behaviour or colour alone
- Each answer must lead to either a new question or a final identification

Working Scientifically — The Essential Skill

'Working Scientifically' questions appear in EVERY FSCE paper. They test whether you can design fair experiments, identify variables, and evaluate evidence. This is often worth more marks than factual recall.

Types of variables

Key Term	Definition
Independent variable	The ONE thing you deliberately CHANGE in an experiment. There must only be ONE.
Dependent variable	What you MEASURE or observe — it depends on what you changed.
Control variables	Everything else you keep the SAME to make the test FAIR.
Fair test	An experiment where only the independent variable is changed; everything else is controlled.
Prediction	A statement of what you expect to happen, based on your scientific knowledge.
Hypothesis	A testable explanation or prediction for a scientific question.
Repeat readings	Carrying out the same test multiple times — reduces the effect of anomalous (odd) results.
Anomalous result	A result that does not fit the pattern — may be due to error; should be investigated.
Conclusion	A statement explaining what your results show, referring back to your prediction.

Worked example — identifying variables:

Experiment: 'Does the amount of water affect how tall a sunflower grows? Six sunflowers are given 0ml, 10ml, 20ml, 30ml, 40ml and 50ml of water daily for four weeks. All other conditions are the same.'

Independent variable: amount of water given each day

Dependent variable: height of the sunflower (what is measured)

Control variables: type of soil, pot size, position (light), temperature, seed type — all kept the SAME

What would make it unfair? If one pot was in the shade, or used different soil — these would be second variables.

Types of scientific enquiry

Type of enquiry	What it involves
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Fair test / Controlled experiment	Change one variable, measure another, control the rest — the most common type
Observation over time	Watch and record how something changes — e.g. watching a caterpillar change into a butterfly
Pattern seeking	Look for relationships in data without doing a controlled test — e.g. comparing arm span and height
Classifying and grouping	Sort objects or organisms into groups based on features
Research using secondary sources	Use books, internet or experts rather than doing a practical

Displaying results

Type of chart/graph	When to use it
Bar chart	When the independent variable is categoric (named groups) — e.g. favourite animals, types of leaf
Line graph	When BOTH variables are numbers and continuous — e.g. temperature over time, height vs. watering
Pie chart	To show proportions of a whole — e.g. percentage of each habitat type
Tally chart	To count occurrences of events — useful in fieldwork
Scatter graph	To look for correlation between two measured variables

10 FSCE-Style Questions — Week 1: Biology

Instructions

Answer all 10 questions. For multiple choice questions, circle the letter of your answer. For short written answer questions, write clearly in the space provided. You will NOT lose marks for wrong answers — always attempt every question.

Q1**Science —
Biology**

Look at this food chain: Grass → Rabbit → Fox → Eagle. The rabbit population suddenly falls because of a disease. What is most likely to happen to the fox population in the short term?

- A. It will increase because there is less competition.
- B. It will decrease because its main food source has reduced.
- C. It will stay exactly the same.
- D. It will increase because eagles eat fewer foxes.

My answer: _____

Q2**Science —
Biology**

Name the life process that describes a living thing removing waste products from its body. (It is one of the letters in MRS GREN.)

Hint: The word begins with E.

Q3**Science —
Working
Scientifically**

Priya wants to test whether fertiliser makes plants grow taller. She plants four identical bean seeds in the same type of soil. She gives each plant a different amount of fertiliser but keeps all other conditions the same. She measures the height of each plant after four weeks. What is the DEPENDENT variable in her experiment?

- A. The type of soil
- B. The amount of fertiliser given to each plant
- C. The height of the plant after four weeks
- D. The length of the experiment (four weeks)

My answer: _____

Q4**Science —
Biology**

A sycamore seed has a wing shape and falls spinning through the air. A burdock seed has tiny hooks all over it. Name the method of seed dispersal for EACH seed:
Sycamore: _____ Burdock: _____

Hint: Think about what the features of each seed are designed for.

Q5**Science —
Biology**

A student finds an animal in the garden. It has six legs, three body sections, wings and an exoskeleton (hard outer casing). Which group does this animal belong to?

- A. Arachnids
- B. Insects
- C. Crustaceans
- D. Worms

My answer: _____

Q6**Science —
Biology**

Which part of a plant absorbs water and minerals from the soil? What ELSE does this part do?

Hint: Think about what the plant needs from the ground — and why it needs to be anchored.

Q7**Science —
Biology**

A butterfly has FOUR stages in its life cycle: egg, larva, pupa, adult. A grasshopper has THREE stages: egg, nymph, adult. What is the main difference between a butterfly's life cycle and a grasshopper's?

- A. The butterfly lays more eggs.
- B. The butterfly goes through complete metamorphosis with a pupa stage; the grasshopper does not have a pupa stage.
- C. The grasshopper takes longer to reach adulthood.
- D. The butterfly's larva looks identical to the adult.

My answer: _____

Q8**Science —
Working
Scientifically**

Tom wants to find out if the colour of light affects how well a plant grows. He sets up three identical plants — one under red light, one under blue light, one under white light. He gives each plant the same amount of water and keeps them at the same temperature. Name ONE control variable in this experiment, and explain why it must be kept the same.

Hint: A control variable is something that could affect the result but is NOT what you are testing.

Q9**Science —
Biology**

In the human digestive system, where is most of the absorption of nutrients into the blood carried out?

- A.** The stomach
- B.** The large intestine
- C.** The small intestine
- D.** The mouth

My answer: _____

Q10**Science —
Biology**

A polar bear has a thick layer of fat under its skin and dense white fur. Explain how EACH of these features helps it survive in its Arctic habitat.

Hint: Think about what the Arctic environment is like — what are the main challenges for survival there?

ANSWER KEY

Week 1 — For Parent / Teacher Use Only

IMPORTANT: Do not give this page to your daughter before she completes the practice questions.

Q1 Science — Food Chains

Answer: B

If the rabbit population falls, the fox loses its main food source and its population will fall too. This is called a knock-on effect through the food chain. Option A is wrong (the foxes have less food, not more); D is wrong (eagles eating fewer foxes is a secondary effect that comes later).

Q2 Short Written Answer

Answer: EXCRETION

Excretion is the removal of waste products made by the body's chemical reactions (e.g. carbon dioxide from respiration, urea in urine). Do not confuse with egestion (removing undigested food as faeces) — they are different processes.

Q3 Science — Working Scientifically

Answer: C

The dependent variable is what is measured — here, the height of the plant. The independent variable (what is changed) is the amount of fertiliser (B). The soil type is a control variable (A). The duration is the time frame, not the dependent variable.

Q4 Short Written Answer

Answer: Wind (sycamore) / Animals — external (burdock)

The sycamore's wing causes it to spin and be carried by wind. The burdock's hooks catch on the fur or clothing of passing animals. Both adaptations ensure seeds travel away from the parent plant.

Q5 Science — Classification

Answer: B

Six legs + three body sections + wings + exoskeleton = insect. Arachnids have 8 legs (spiders, mites). Crustaceans have more than 6 legs and usually live in water (crabs, woodlice). Worms have no legs at all.

Q6 Short Written Answer

Answer: Roots — absorb water/minerals AND anchor the plant

Roots have two main functions: absorbing water and dissolved minerals from the soil, and anchoring the plant so it stays upright. This is a two-part answer — award one mark for each function.

Q7 Science — Life Cycles

Answer: B

The key difference is the pupa stage. Butterfly = complete metamorphosis (4 stages: egg → larva → pupa → adult). Grasshopper = incomplete metamorphosis (3 stages: egg → nymph → adult). The nymph looks like a miniature adult but has no wings.

Q8 Short Written Answer

Answer: Accept: amount of water given / temperature / pot size / type of soil / position of plant — with explanation

A correct answer names ONE variable AND explains that if it changed, it could affect plant growth and make it impossible to know whether the light colour caused the difference. Example: 'The amount of water must stay the same because if one plant got more water it might grow better for that reason, not because of the light colour.'

Q9 Science — Human Digestive System

Answer: C

Most nutrient absorption happens in the small intestine, which has a very large surface area (covered in tiny finger-like projections called villi — KS3 extension but useful to know). The large intestine mainly absorbs water. The stomach breaks down food but absorption is minimal there.

Q10 Short Written Answer

Answer: Fat = insulation from cold / White fur = camouflage in snow

Fat (blubber): acts as insulation — traps heat and prevents body heat from escaping in the freezing Arctic temperatures. Also provides energy reserves when food is scarce. White fur: provides camouflage against the snow, making the bear less visible to prey and helping it hunt. Accept other valid points such as dense fur trapping air for warmth.

Week 1 Score: _____ / 10

Subjects to revisit next session (tick any that scored 0 or were unclear):

- ☐ Food chains and food webs (Q1, Q10)
- ☐ MRS GREN / life processes (Q2)
- ☐ Fair testing and variables (Q3, Q8)
- ☐ Seed dispersal (Q4)
- ☐ Classification / vertebrates (Q5)
- ☐ Plant structure and functions (Q6)
- ☐ Life cycles and metamorphosis (Q7)
- ☐ Human digestive system (Q9)

Coming up: Week 2 — Science: Chemistry (States of Matter · Materials · Mixtures · Reversible & Irreversible Changes)